

Tide retrospective:

Harmonic GibbsObservable instances for monomials

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7 May 2026

Abstract

A single Tide-loop excursion (**G4** from the 7 May candidates survey) closing the three carried `Threepoint.GibbsObservable` hypotheses on Tide 13’s harmonic-cross-susceptibility-derivative theorem. Headline: `Threepoint.harmonic_id_gibbsObservable.pow` for any $k : \mathbb{N}$, proved by dominated differentiation under the integral with a Gaussian-shaped dominator from Young’s inequality. 397 lines of Lean; zero `sorry`, `axiom`, `native.decide`; clean diagnostics. The session’s load-bearing event was a GPT-5.5 Pro consult that pivoted the dominator-integrability argument from a piecewise $|x| \leq 1 / |x| > 1$ split to a single global `rpow-cast` route, and supplied the `nlinarith` hints $((\lambda x \pm 2h)^2 \geq 0)$ for Young after a first attempt accumulated seven distinct frictions across a 260-line draft.

1 Setting

This Tide is the natural close-out of Tide 13, which proved $\partial_h \text{Cov}_h(x, x)|_{h=0} = 0$ for the harmonic Gibbs measure but had to carry three `GibbsObservable` hypotheses (for x, x^2, x^3) as parameters because the harmonic monomial `GibbsObservable` instances weren’t yet in the seabed. G4 supplies them, generic in $k : \mathbb{N}$, making the Tide 13 theorem unconditional for the canonical monomial cases.

The mathematical content is mild: for the harmonic potential $L(x) = \lambda x^2/2$ with linear perturbation $A(x) = x$, the perturbed numerator $N_k(h) := \int x^k \exp(-t(L(x) + hA(x))) dx$ is differentiable in h at $h = 0$ with derivative $\int x^k \cdot (-t \cdot x) \cdot \exp(-tL(x)) dx$. This is dominated differentiation under the integral, valid because the perturbed integrand has a uniform Gaussian-shaped dominator on $|h| \leq 1^1$.

The Step 2 deliberation (Claude $\leftrightarrow$ GPT-5.5 Pro) drafted five candidates (a finite- k closed-form route, a generic- k closed-form route, the dominated-differentiation route, a global-Cov constancy result, and a ‘*just close G4*’-shaped variant). Both parties backed the dominated-differentiation route (Candidate D) — generic in k at no extra cost, no Gaussian-moment library required.

2 The thing formalised

For $\lambda > 0$, $t > 0$, and any $k \in \mathbb{N}$, the canonical monomial $(x \mapsto x^k)$ is a `GibbsObservable` for the harmonic potential $(\lambda/2)x^2$ with linear perturbation $A(x) = x$ under Lebesgue measure on $\mathbb{R}$: the perturbed numerator $\int x^k \cdot \exp(-t(L(x) + hA(x))) dx$ reduces to the unperturbed integral at $h = 0$, and is differentiable at $h = 0$ with the expected pointwise-derivative integral.

¹Young’s inequality: $|h \cdot x| \leq (\lambda/4)x^2 + h^2/\lambda$ absorbs the linear perturbation into the quadratic decay.

The Lean signature:

```
theorem Threepoint.harmonic_id_gibbsObservable_pow
  {lam t : ℝ} (hlam : 0 < lam) (ht : 0 < t) (k : ℕ) :
  Threepoint.GibbsObservable (volume : Measure ℝ)
    (fun x : ℝ => lam / 2 * x ^ 2)
    (fun x : ℝ => x) t (fun x : ℝ => x ^ k)
```

located in `Laplace/OneD/HarmonicGibbsObservableMonomials.lean`, namespace `Laplace.OneD`.

The result is the natural pair to Tide 10's harmonic GibbsRegularity instance: where the GibbsRegularity instance handles the *partition-function* differentiability uniformly in φ (via the closed-form route), this GibbsObservable instance handles the *numerator* differentiability per-observable. Together they discharge the analytic content of the harmonic FDT and cross-susceptibility theorems, instantiated at any monomial.

3 Proof strategy

The Lean proof is built from five small lemmas, each with a clean mathematical narrative:

Young's inequality with weight $\lambda/2$. For $\lambda > 0$, $|h \cdot x| \leq (\lambda/4)x^2 + h^2/\lambda$. The two one-sided bounds come from $(\lambda x - 2h)^2 \geq 0$ and $(\lambda x + 2h)^2 \geq 0$ after dividing through by 4λ . The Lean idiom that landed cleanly: clear the $/\lambda$ by `field.simp + ring`, then `nlinarith` with the square hint.

Dominator integrability. $\int |x|^k \exp(-cx^2) dx < \infty$ for $c > 0$, $k \in \mathbb{N}$. Lean route: get $\int x^{(k:\mathbb{R})} \exp(-cx^2) dx < \infty$ from `integrable_rpow_mul_exp_neg_mul_sq` (Mathlib's real-power Gaussian moment), cast to `Nat-power` via `Real.rpow_natCast`, then bridge x^k to $|x|^k$ via `Integrable.norm` (since $\|x^k \cdot e^{-cx^2}\| = |x|^k \cdot e^{-cx^2}$ because the exponential is positive).

Pointwise integrand bound. $\|x^k \exp(-t(\lambda/2 \cdot x^2 + hx))\| \leq e^{t/\lambda} \cdot |x|^k \cdot e^{-(t\lambda/4)x^2}$ for $|h| \leq 1$. Direct application of Young followed by `exp-monotonicity`.

Pointwise differentiability. For each fixed x , the function $h \mapsto x^k \cdot \exp(-t(\lambda/2 \cdot x^2 + hx))$ is differentiable in h by the standard `HasDerivAt.exp.const_mul` chain. The derivative is $x^k \cdot (-(t \cdot x)) \cdot \exp(-t(\lambda/2 \cdot x^2 + hx))$.

Headline assembly via dominated differentiation. Apply the parametric-integral derivative lemma² with the four hypothesis families: (i) the local neighbourhood `Metric.ball 0 1 ∈ nhds 0`; (ii) AE strong measurability via `fun_prop`; (iii) integrability of the derivative integrand bounded by the dominator at index $k + 1$ (the derivative integrand is $-t \cdot x^{k+1} \cdot \exp(-t(\dots))$); (iv) pointwise `HasDerivAt` from the chain rule above. The result's second component is the desired `HasDerivAt`; a small `integral_congr_ae` massages $0 \cdot x = 0$ inside the perturbed integrand to match the GibbsObservable shape.

²`MeasureTheory.hasDerivAt.integral_of_dominated_loc_of_deriv_le`.

4 Roadblocks and resolutions

The first attempt accumulated seven distinct errors. I drafted all the analytic-core lemmas in one pass (~260 lines) and tried to compile. The build report showed three `nlinarith` failures (Young’s inequality, AM-GM bounds), two `Real.rpow` type-mismatches (the dominator integrability), one `mul.le.mul` argument-order issue, and one bound-shape mismatch. Each error was individually fixable but the combined diagnosis cost would have been slow. The lean-formalisation skill’s *50% probability of direct proof $\implies$ factor first* rule applied; I reverted and wrote the handoff note while launching a focused GPT-5.5 Pro consult.

The GPT consult was decisive. Three pieces of advice landed:

GPT-5.5 Pro: For Young: don’t ask `nlinarith` to reason directly with `/ lam`. Multiply by 4λ first, then `nlinarith` with $(\lambda x \mp 2h)^2 \geq 0$.

GPT-5.5 Pro: For dominator integrability: there’s a *global* dominator $M \cdot e^{-(c/2)x^2}$, $M = e^{k^2/(2c)}$, via the same Young applied to $(k, |x|)$. No piecewise $|x| \leq 1$ / $|x| > 1$ split needed.

GPT-5.5 Pro: For `HasDerivAt`: the canonical chain is
`(((((const).add (id.mul_const)).const_mul).neg).exp).const_mul`.

With these in hand the second attempt landed clean.

In the end, the dominator integrability route I used differed from GPT’s advice: rather than the global Young-based AM-GM dominator, I used `Integrable.norm` on `integrable_rpow_mul_exp_neg_mul_sq` at `s = (k:ℝ)`, which is even simpler. (~10 lines vs ~50 for the AM-GM route.) Worth recording — both routes work; the Mathlib `Integrable.norm` bridge between x^k and $|x|^k$ is ergonomically excellent for this case.

Pi.add unfolding edge-case. The `HasDerivAt.add` of `(fun _ => c)` and `(fun y => y * x)` produces a goal in `((fun _ => c) + (fun y => y * x))` form (with `Pi.add`), not the inline-lambda form `(fun y => c + y * x)`. `simp` normalised $0 + x$ inside the goal too aggressively. Resolved by introducing a manual `hzero : (0 : ℝ) + x = x = zero.add x` and `rwa`-ing it into the `hsum` term *before* using.

Continuity proofs via fun_prop. Manual continuity chains³ hit the same `Pi.add` boundary as the `HasDerivAt` issue. Switching to `by fun_prop` sidestepped this entirely. Worth a `CLAUDE.md` note for the next dominated-differentiation Tide.

5 What was Mathlib, what was new

Mathlib provided. `MeasureTheory.hasDerivAt.integral_of_dominated_loc_of_deriv_le` (the main parametric-integral lemma); `integrable_rpow_mul_exp_neg_mul_sq` (real-power Gaussian moment integrability); the `Integrable.norm` / `Real.rpow_natCast` bridge; the `HasDerivAt.exp`, `.const_mul`, `.add`, `.neg`, `.mul_const` combinators; `Real.exp_le_exp.mpr`, `Real.exp_add`, `abs_le`, `abs_pow`, `Real.norm_eq_abs`.

³Of the form `Continuous.const_mul.add.const_mul.neg.exp`.

The seabed provided. `Threepoint.GibbsObservable` (the target structure) and Tide 13’s `cov_h_id_id_deriv_harmonic_eq_zero` (the consumer of this Tide’s output). No prior 1D Gaussian moment infrastructure was used directly; the proof routes around the seabed by going straight through Mathlib’s parametric-integral API.

What was new in this tide. The Young’s-with-weight inequality (likely promotable to `lean-common`); the dominator integrability lemma; the pointwise integrand bound; the pointwise `HasDerivAt` for the perturbed monomial integrand; the headline `GibbsObservable` instance.

6 Lessons

- *Strategic GPT consults at the friction point pay off asymmetrically.* The first attempt accumulated seven distinct errors in a 260-line draft; the GPT consult turned them into a clean second pass. The pattern matches what the lean-formalisation skill describes as the ~ 5 -minute consult that saves a multi-hour detour.
- *`nlinarith` can’t divide.* For inequalities with `/ c`, multiply through first via `field_simp + ring`, then invoke `nlinarith` with the right square hint. Generic enough to lift to `laplace/CLAUDE.md`.
- *`fun_prop` for continuity beats manual chains.* The continuity proofs were the second-most-fragile part of the first attempt. `fun_prop` dispatched all of them in one tactic call.
- *`Integrable.norm` is the right bridge between x^k and $|x|^k$.* Cleaner than piecewise $|x| \leq 1 / |x| > 1$ splits when the inner integrability already exists in Mathlib for x^k . One more pattern for the API reference.

7 Follow-ups

- *Promote `abs_mul_le_quarter_lambda_sq_add` to `lean-common`.* Standard Young’s-with-weight inequality; if a second use materialises (e.g. in `singfluence-lean` or in a future bounded-prior Tide), worth the promotion.
- *Package the dominated-differentiation infrastructure as a reusable `GibbsObservable` factory.* The four hypothesis discharges (integrability of F at $h = 0$, AE strong measurability, integrand bound, pointwise differentiability) are generic for any φ with a Gaussian-shaped dominator. A future Tide could extract `Threepoint.gibbsObservable_of_dominated` so that each new observable family doesn’t re-run the whole hypothesis chain.
- *Tide 13 unconditionalisation.* With G4 closed, the next natural Tide is to specialise Tide 13’s `cov_h_id_id_deriv_harmonic_eq_zero` to the unconditional case for $\varphi(x) = x$ and $B(x) = x$ by substituting in the new instances. ~ 20 – 30 lines on top of this Tide.
- *Candidate E (deferred at deliberation).* Global covariance constancy: for the harmonic Gibbs measure with linear perturbation, $\text{Cov}_h(x, x)$ equals the unperturbed value $1/(\lambda t)$ for *all* h , not merely vanishing derivative at 0. Different theorem statement; closed-form route via Tide 10’s $Z(h)$; ~ 100 lines.

Acknowledgements

GPT-5.5 Pro contributed the deliberation-stage advisory at Step 2 (picking Candidate D over the closed-form-binomial route), and a mid-formalisation tactical consult that pivoted the analytic-core proof from a friction-stuck draft to a clean landing.⁴

⁴Tide log: `sri/projects/patterning/tide-log/2026-05-07-tide-harmonic-gibbsobservable-monomials.md`.
Tactical consult on Lean frictions: `sri/projects/patterning/tide-log/gpt55_g4_frictions_v1.md`.